

IT3051 - FUNDAMENTALS OF DATA MINING
FINAL TECHNICAL REPORT



**Air Quality Risk Classification Using EPA Historical
Monitoring Data**

Next-Year PM2.5 Risk Category Prediction

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Executive Summary

This project develops an end-to-end data mining system that predicts the next-year PM2.5 air-quality risk category for a U.S. EPA monitoring site using historical annual monitoring statistics and contextual site information. The work follows the complete workflow required by IT3051: problem definition, dataset validation, data engineering, exploratory analysis, leakage-safe preprocessing, model development, hyperparameter optimization, final model selection, backend/frontend integration, and testing.

The original EPA Air Quality Annual Summary data contained 2,038,710 rows and 55 columns. After selecting PM2.5 parameter code 88101, cleaning, site-year aggregation, target engineering, and historical feature engineering, the model-ready dataset contained 16,535 records from 1,849 monitoring sites covering 1997–2016. The target was strongly imbalanced; the Hazardous class had only four observations and was merged with Very Unhealthy, producing five final classes.

A chronological split was used to preserve the forecasting nature of the task: 1997–2012 for training (13,191 records), 2013–2014 for validation (1,783), and 2015–2016 for testing (1,561). Preprocessing was fitted only on training data. Five algorithms were compared: Multinomial Logistic Regression, Random Forest, Gradient Boosting, LightGBM, and RBF Support Vector Machine (SVM). Macro-F1 was the primary model-selection metric because it gives equal importance to minority and majority risk classes.

The tuned RBF SVM was selected with $C=5.0$, $\text{gamma}=\text{'scale'}$, and $\text{class_weight}=\text{'balanced'}$. It achieved validation Macro-F1=0.4364 and accuracy=75.49%, outperforming the next-best Random Forest by 0.0273 Macro-F1. On the untouched test set, it achieved 79.24% accuracy, 0.2945 Macro-F1, 0.2968 balanced accuracy, and QWK=0.4014. The difference between accuracy and Macro-F1 highlights the continuing effect of class imbalance and temporal distribution shift.

The selected pipeline was integrated into a local web application. The backend validates inputs, loads the saved model, applies the same feature engineering and preprocessing, supports single and batch predictions, and returns category-specific guidance. Automated backend tests passed 37/37 in the final verification environment. The system is an academic decision-support prototype and is not an official EPA forecast or public-health advisory.

1. Introduction and Problem Definition

1.1 Background

Fine particulate matter (PM2.5) is an important air-quality indicator because elevated concentrations are associated with adverse respiratory and cardiovascular outcomes. EPA monitoring stations provide long-term measurements that can be transformed into a supervised learning problem. Raw annual statistics, however, are difficult for non-specialist users to interpret directly. This project therefore translates historical monitoring patterns into a categorical next-year risk estimate.

1.2 Problem Statement

Given a monitoring site's historical annual PM2.5 statistics and contextual information, the system predicts the likely PM2.5 risk category for the following year. Only information available before the prediction year is used. This design prevents look-ahead leakage and makes evaluation closer to a realistic forecasting scenario.

1.3 Prediction Objective

The primary objective is supervised multi-class classification of the engineered target `pm25_risk_category_next_year`. The deployed system additionally presents a model confidence estimate when supported by the saved estimator, a plain-language category explanation, and activity guidance based on a user-selected sensitivity threshold.

1.4 Scope

The system is an academic annual-planning prototype. It predicts project-defined PM2.5 risk categories from historical annual summary data; it does not generate an official AQI forecast, replace EPA advisories, or provide a causal explanation of pollution.

2. Scenario, Stakeholders and Requirements

Stakeholder	Potential Use
Public health departments	Prioritize sites that may warrant closer monitoring or communication.
Local environmental agencies	Identify locations trending toward higher annual PM2.5 risk.
Schools and care facilities	Use more precautionary activity thresholds for sensitive populations.
Urban planners / policy makers	Review longer-term spatial and temporal risk patterns.
General public / advisory users	Receive understandable risk-category and activity guidance.

2.1 Functional Requirements

- Accept annual monitoring-site measurements and contextual location information.
- Predict one of five next-year PM2.5 project risk categories.
- Validate numeric and textual inputs and return understandable error messages.
- Allow selection of an existing EPA site-year or manual data entry.
- Support batch CSV prediction for up to 200 site rows.
- Provide configurable activity-alert sensitivity without changing the model prediction.
- Return a confidence estimate only when the saved model supports probability output.

2.2 Non-Functional Requirements

- Usable by non-specialist staff.
- Consistent preprocessing between training and deployment.
- Reasonable response time for annual/seasonal planning.
- Robust handling of missing historical information.
- Reproducible code structure and documented model evaluation.
- Clear warning that outputs are decision-support estimates, not official advisories.

3. Dataset Identification and Understanding

3.1 Source

The project uses the Kaggle 'EPA Air Quality' dataset, derived from the U.S. Environmental Protection Agency Air Quality System (AQS). The original annual-summary data covers multiple pollutants and monitoring configurations. The project focuses on PM2.5 mass concentration under local conditions (EPA parameter code 88101).

Item	Project Value
Original dataset size	2,038,710 rows × 55 columns
Model-ready dataset	16,535 rows × 38 columns
Unique monitoring sites	1,849
Usable model-ready years	1997–2016
Target	pm25_risk_category_next_year
Task	Supervised multi-class classification
Original target classes	6
Final target classes	5 after rare-class merge

3.2 Target Engineering

The target is not a field supplied directly by the original dataset. It was engineered from the following year's PM2.5 information using documented project breakpoints. The next-year value is shifted within each monitoring site so that current/past predictors are paired with a future category. Target-year measurements are never used as predictors.

3.3 Original Class Distribution

Class	Records	Percent
Good	86	0.52%
Moderate	9,418	56.96%
Unhealthy for Sensitive Groups	5,578	33.73%
Unhealthy	1,398	8.45%
Very Unhealthy	51	0.31%
Hazardous	4	0.02%

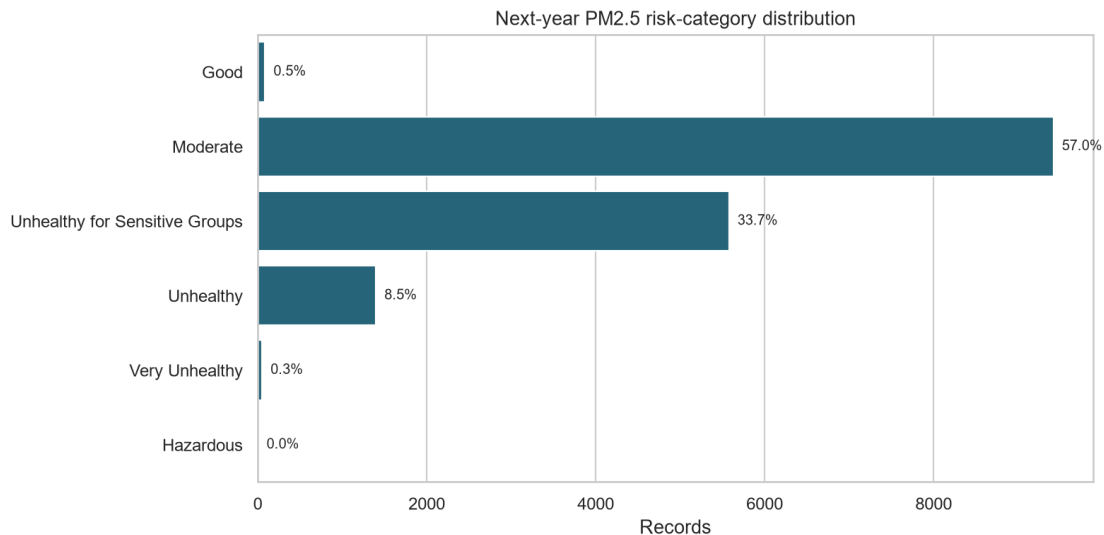


Figure 1. Original target-class distribution.

The distribution is highly imbalanced. Because Hazardous had only four records, it could not be represented reliably across chronological training, validation, and test periods. It was therefore merged into Very Unhealthy before modelling.

4. Data Cleaning and Preparation

4.1 PM2.5 Extraction and Site-Year Aggregation

The data-engineering pipeline filters the raw file to parameter code 88101, normalizes state/county/site identifiers, creates a stable site_id, converts relevant measurements to numeric form, validates geographic and monitoring

ranges, resolves multiple monitoring configurations, and aggregates records to a single monitoring-site/year representation.

4.2 Data Quality Checks

Check	Result / Action
Exact duplicate rows	0; duplicate handling retained in pipeline.
Duplicate site-year records	0 in ML-ready dataset.
Standard missing values	0 detected before textual-marker conversion.
Textual invalid values	'unknown' markers converted to NaN before imputation.
Infinite numeric values	0
Missing target values	0
Outliers	Investigated with IQR diagnostics; extreme pollution values were not removed automatically because they may represent genuine events.

4.3 Leakage Prevention

Leakage control was treated as a core requirement. Historical lags and rolling features were created within each site using past observations only. The target was removed from predictors. The split was chronological, and all fitted preprocessing operations were learned from training data only before being applied unchanged to validation and test data.

5. Exploratory Data Analysis

EDA examined class imbalance, temporal patterns, geographic variation, feature distributions, outliers, missing/unknown markers, correlations, mutual information, and potential leakage. No direct leakage candidates were reported in the processed EDA checks.

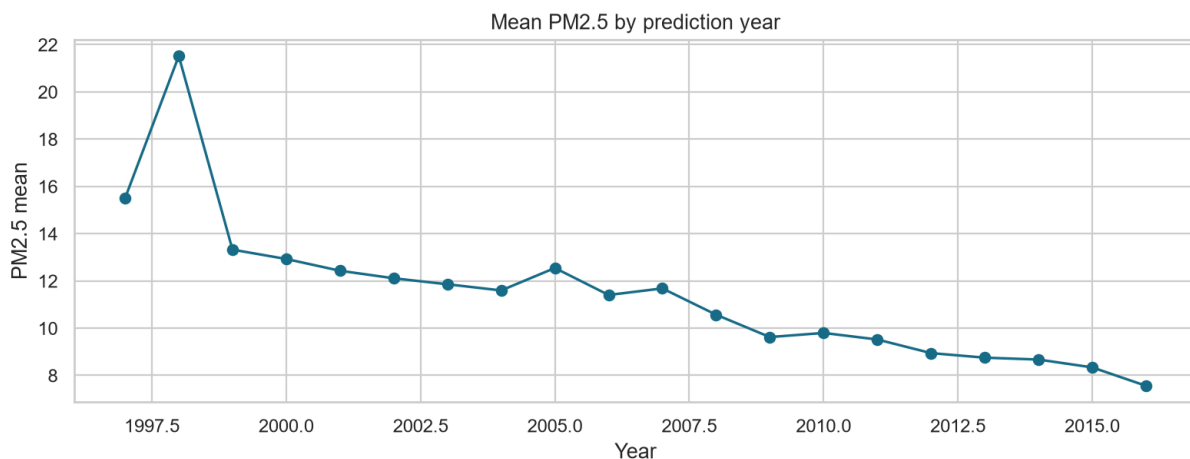
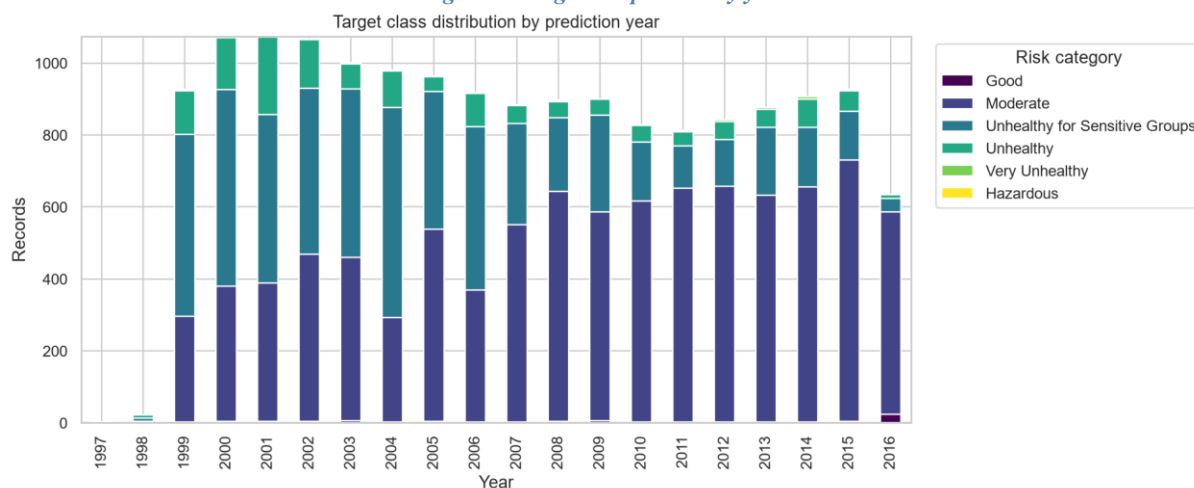


Figure 2. PM2.5 mean across years, showing temporal change.

Figure 3. Target composition by year.



5.1 Feature Relevance

Mutual-information analysis ranked high-percentile pollution measurements and variability/location variables among the most informative predictors. The top features included ninety_five_percentile, arithmetic_standard_dev, longitude, ninety_nine_percentile, ninety_eight_percentile, ninety_percentile, latitude, p95_minus_median, p99_p10_range, first_max_value, and pm25_mean.

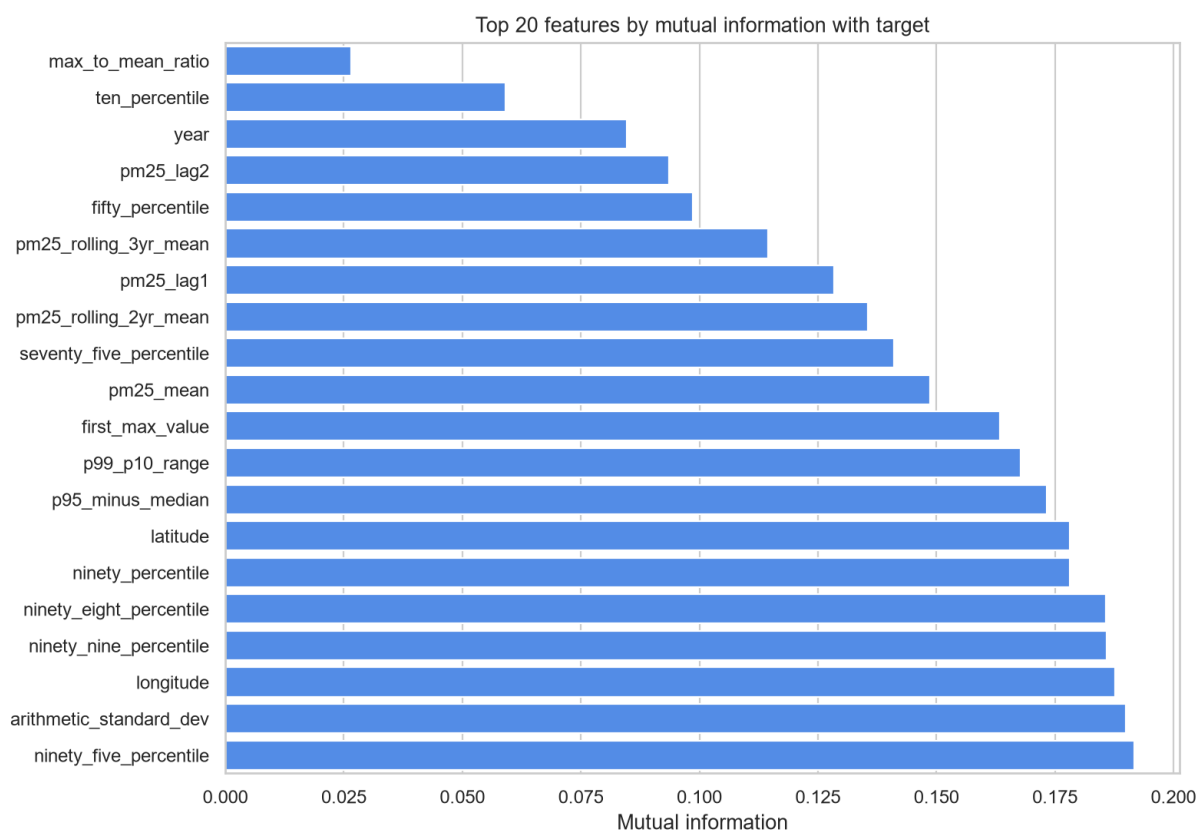


Figure 4. Mutual information between model features and the target.

5.2 Correlation and Multicollinearity

Several annual concentration summaries are naturally correlated because they describe the same site's distribution at different percentiles. Correlation diagnostics were therefore used to identify redundant relationships. Tree-based models can tolerate correlated predictors better than linear models, while regularization/scaling and validation performance were considered for the linear and margin-based alternatives.

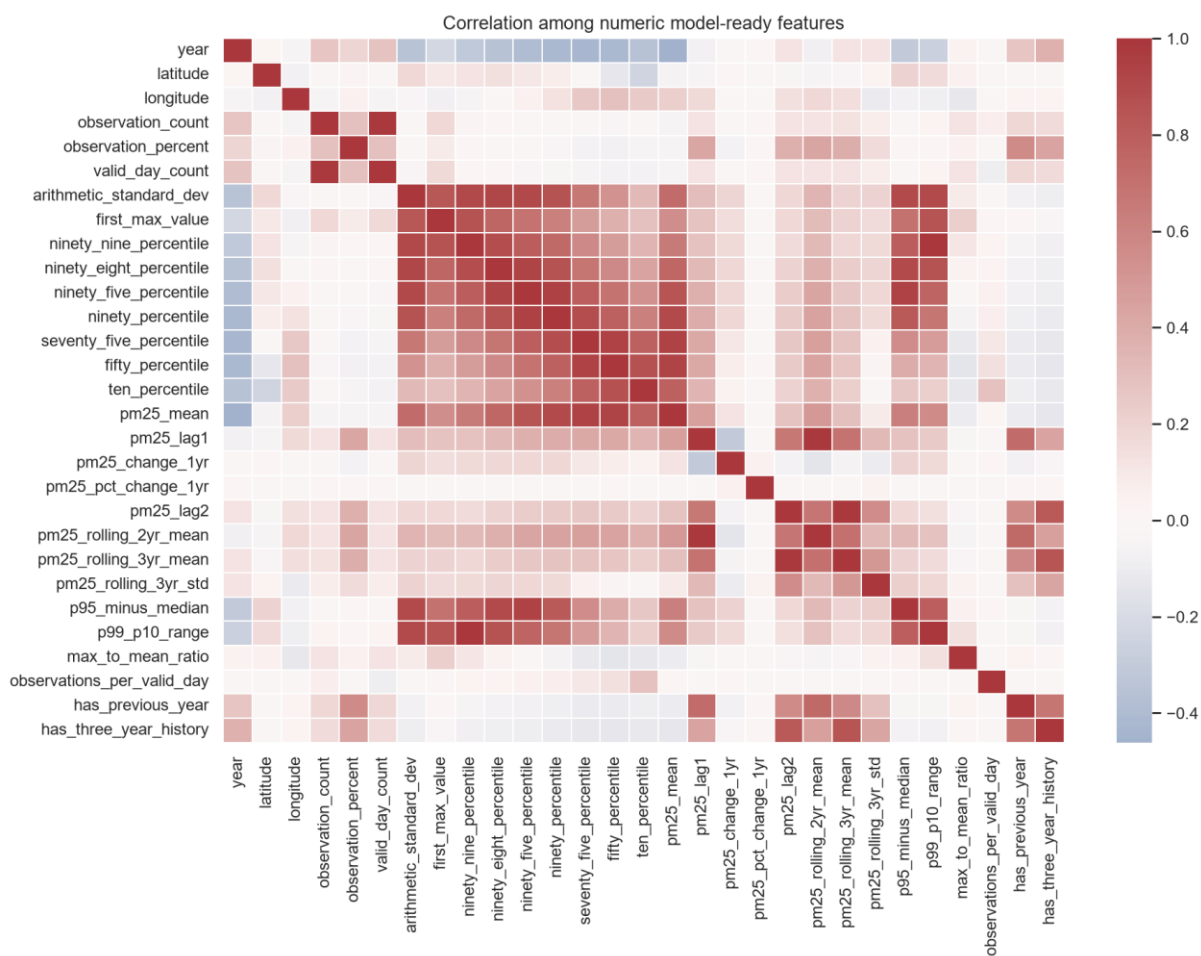


Figure 5. Feature-correlation heatmap.

6. Feature Engineering and Selection

Feature engineering was designed to represent level, variability, recent history, trend, spread, and data availability without using future information.

Feature Group	Examples / Purpose
Current-year pollution	pm25_mean, percentiles, first_max_value, arithmetic_standard_dev
Historical lags	pm25_lag1, pm25_lag2
Trend / change	pm25_change_1yr, pm25_pct_change_1yr
Rolling history	pm25_rolling_2yr_mean, pm25_rolling_3yr_mean, pm25_rolling_3yr_std
Distribution spread	p95_minus_median, p99_p10_range, max_to_mean_ratio
Monitoring intensity	observation_count, observation_percent, valid_day_count, observations_per_valid_day
History availability	has_previous_year, has_three_year_history
Geographic/context	latitude, longitude, state_name, county_name, city_name, cbsa_name

Direct identifiers such as site_id, state_code, county_code, and site_num were excluded from ordinary model predictors. The deployed preprocessing configuration uses 29 numerical predictors and four categorical predictors.

7. Dataset Splitting and Preprocessing

7.1 Chronological Split

Partition	Years	Records
Training	1997–2012	13,191
Validation	2013–2014	1,783
Testing	2015–2016	1,561
Total	1997–2016	16,535

A chronological split was chosen instead of a random split because the application predicts future-year risk. The test set was kept untouched until after final model selection.

7.2 Preprocessing Pipeline

Feature Type	Pipeline
Numerical	Textual 'unknown' → NaN → median imputation → StandardScaler
Categorical	Most-frequent imputation → OneHotEncoder(handle_unknown='ignore')

The ColumnTransformer-based preprocessing pipeline was fitted on training data only. This is especially important for median imputation, scaling, and categorical encoding, because fitting them on later years would leak information from the validation/test periods.

8. Model Development

The assignment requires at least four suitable algorithms. Five were implemented to provide a broader comparison: Multinomial Logistic Regression, Random Forest, Gradient Boosting, LightGBM, and SVM with an RBF kernel. A dummy classifier was also used as a reference baseline in early comparison.

Algorithm	Reason for Inclusion
Multinomial Logistic Regression	Interpretable linear baseline and probability-based multiclass reference.
Random Forest	Captures nonlinear interactions and is robust on tabular data.
Gradient Boosting	Sequentially improves weak learners and models nonlinear structured data.
LightGBM	Efficient gradient-boosted trees suitable for tabular data and tuning.
SVM (RBF)	Nonlinear margin-based classifier; scaling and class weighting can support imbalanced classes.

8.1 Evaluation Metrics

Macro-F1 was the primary selection metric because each class contributes equally, which is important when severe categories are rare. Accuracy, balanced accuracy, macro precision/recall, weighted F1, confusion matrices, and Quadratic Weighted Kappa (QWK) were also examined. Accuracy was treated as secondary because a majority-class-dominated model can appear strong while performing poorly on minority classes.

9. Hyperparameter Tuning and Optimization

Model	Selected Tuned Parameters	Validation Macro-F1
Random Forest	n_estimators=200; max_depth=20; min_samples_leaf=1; class_weight=balanced	0.4091
Gradient Boosting	n_estimators=200; learning_rate=0.1; max_depth=3; subsample=0.8	0.3884
LightGBM	n_estimators=200; learning_rate=0.05; num_leaves=31; max_depth=5; subsample=0.8; colsample_bytree=1.0	0.3825
SVM (RBF)	C=5.0; gamma=scale; class_weight=balanced	0.4364

Tuning improved Macro-F1 for all four tuned nonlinear models. The largest improvement was observed for SVM: from 0.2949 to 0.4364 (+0.1414). Random Forest improved by +0.0631, LightGBM by +0.0618, and Gradient Boosting by +0.0451. This supports the use of hyperparameter optimization rather than selecting models from baseline accuracy alone.

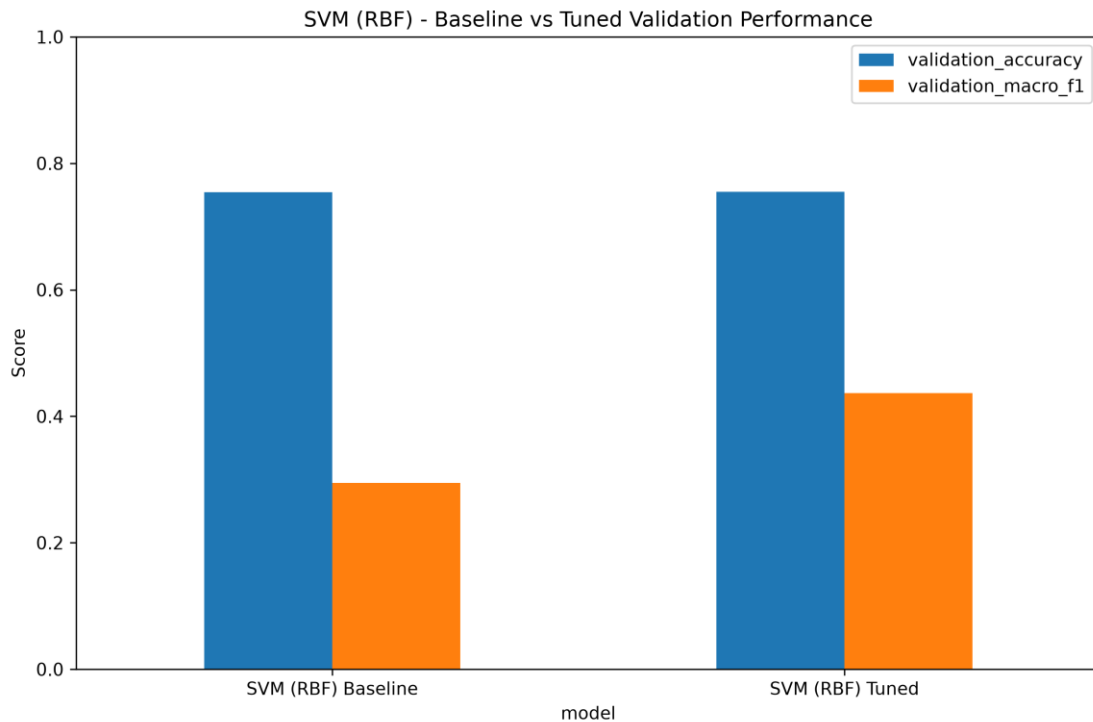


Figure 6. RBF SVM baseline versus tuned validation performance.

10. Final Model Selection and Evaluation

10.1 Validation Comparison

Rank	Model	Macro-F1	QWK	Balanced Acc.	Accuracy
1	SVM (RBF)	0.4364	0.4850	0.4300	0.7549
2	Random Forest	0.4091	0.5165	0.4172	0.7650
3	Gradient Boosting	0.3884	0.4827	0.3774	0.7650
4	LightGBM	0.3825	0.4615	0.3404	0.7656
5	Logistic Regression	0.3485	0.4340	0.4629	0.7213

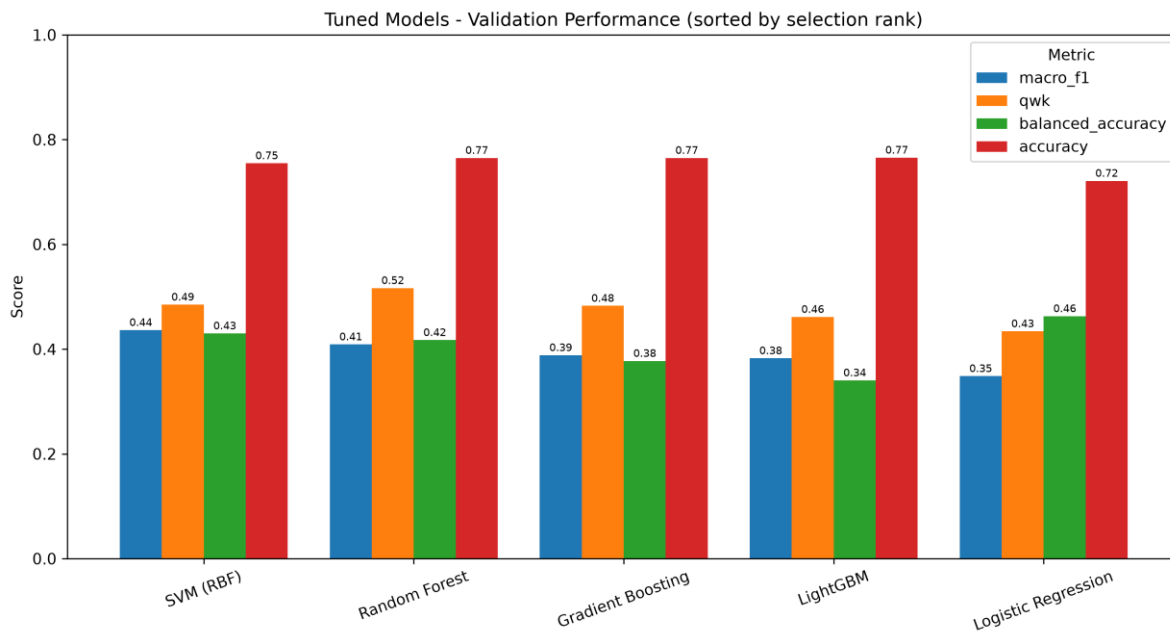


Figure 7. Tuned-model validation comparison.

The final selection rule ranked models by validation Macro-F1, with QWK, balanced accuracy, and accuracy as tie-breakers. SVM achieved the highest Macro-F1 and led the runner-up Random Forest by 0.0273. Therefore, the tuned RBF SVM was selected before the test set was evaluated.

10.2 Final Test Results

Metric	Test Result
Accuracy	0.7924 (79.24%)
Balanced accuracy	0.2968
Macro precision	0.3593
Macro recall	0.2968
Macro F1	0.2945
Weighted F1	0.7786
Quadratic Weighted Kappa	0.4014

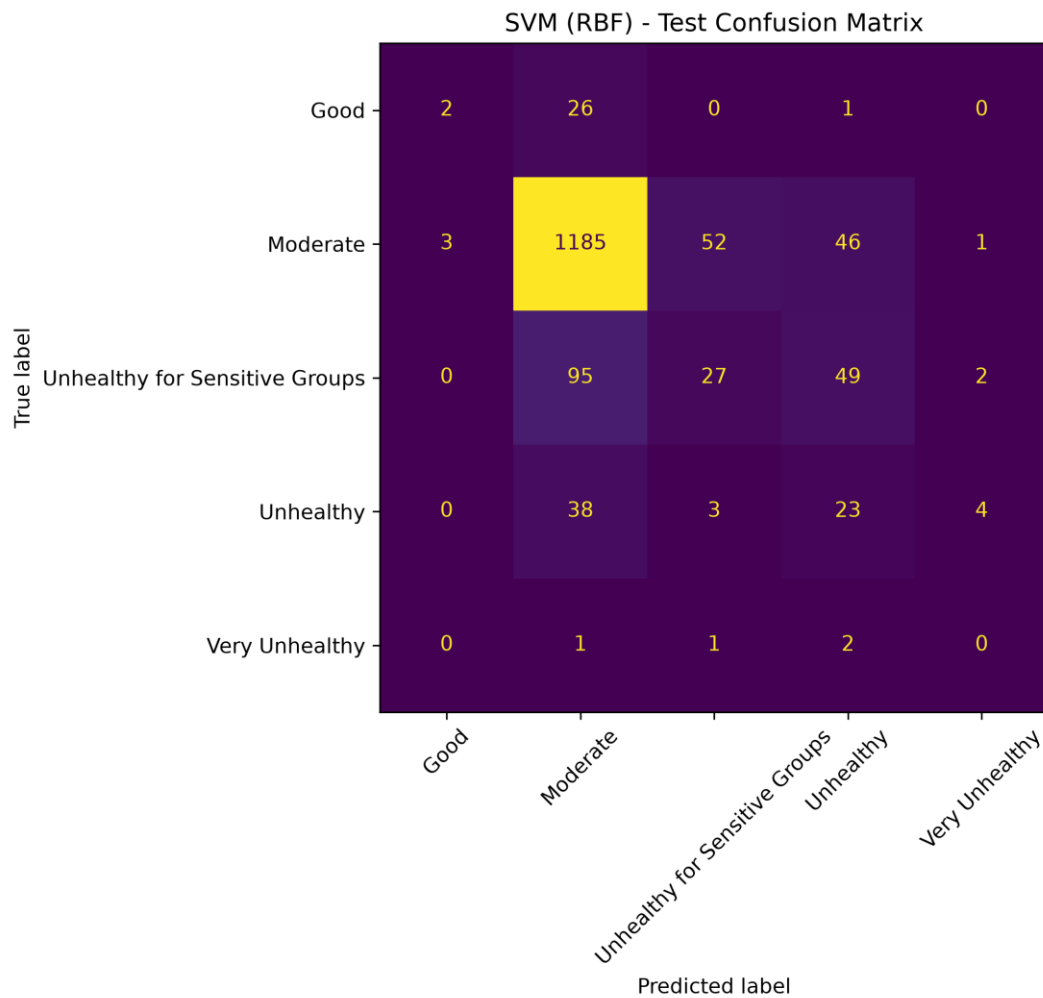


Figure 8. Final SVM confusion matrix on the untouched 2015–2016 test set.

10.3 Per-Class Interpretation

Class	Precision	Recall	F1	Support
Good	0.400	0.069	0.118	29
Moderate	0.881	0.921	0.900	1,287
Unhealthy for Sensitive Groups	0.325	0.156	0.211	173
Unhealthy	0.190	0.338	0.243	68
Very Unhealthy	0.000	0.000	0.000	4

The final model performs strongly on the dominant Moderate class but remains weak on rare categories. This explains why overall accuracy is high while Macro-F1 and balanced accuracy are much lower. The result should therefore be interpreted as evidence of a usable academic prototype with substantial minority-class limitations, not as production-ready public-health performance.

11. System Architecture and Implementation

The implemented system follows a simple layered architecture:

Layer	Main Components	Responsibility
Data layer	EPA annual summary → interim/model-ready CSV	Stores historical monitoring records and engineered data.
Feature layer	src/features/feature_engineering.py	Creates history/trend/spread features consistently.
Model pipeline	models/best_model.joblib	Contains fitted preprocessing and selected SVM classifier.
Backend	app/backend/server.py	Validates requests, loads model, builds features, predicts, supports batch CSV.
Frontend	HTML/CSS/JavaScript	Collects inputs, displays measurement profile, risk category, confidence/guidance.
Testing	tests/test_prediction_app.py	Checks validation, feature schema, single/batch predictions and HTTP endpoints.

End-to-end flow: User input or selected EPA site-year → backend validation → feature construction → saved preprocessing/model pipeline → predicted category/probability → category explanation and activity guidance → frontend result display.

12. Backend and Frontend Development

12.1 Backend

The backend is implemented with Python's threaded HTTP server. It serves the frontend and exposes prediction endpoints. Numeric inputs are checked for finite values and domain-specific ranges; text inputs are required and length-limited. The backend loads `models/best_model.joblib`, checks the prediction feature schema, and uses the same feature-engineering code as model development.

The application also supports batch CSV predictions with a maximum of 200 rows. Invalid batch rows are reported individually rather than causing all valid rows to fail.

12.2 Frontend and User Experience

The frontend provides cascading location/site/year selection using the model-ready EPA data and also supports manual entry. It groups fields into location, air-quality measurements, monitoring information, and results. Users can select General Public, Sensitive Groups, or High Precaution alert settings. These thresholds affect activity guidance only; they do not alter the model's predicted category.

The interface includes a downloadable batch CSV template and returns downloadable batch results. This supports both single-site exploration and a regional overview use case.

12.3 Frontend Screenshots and System Demonstration

The following screenshots were captured from the successfully running web application.

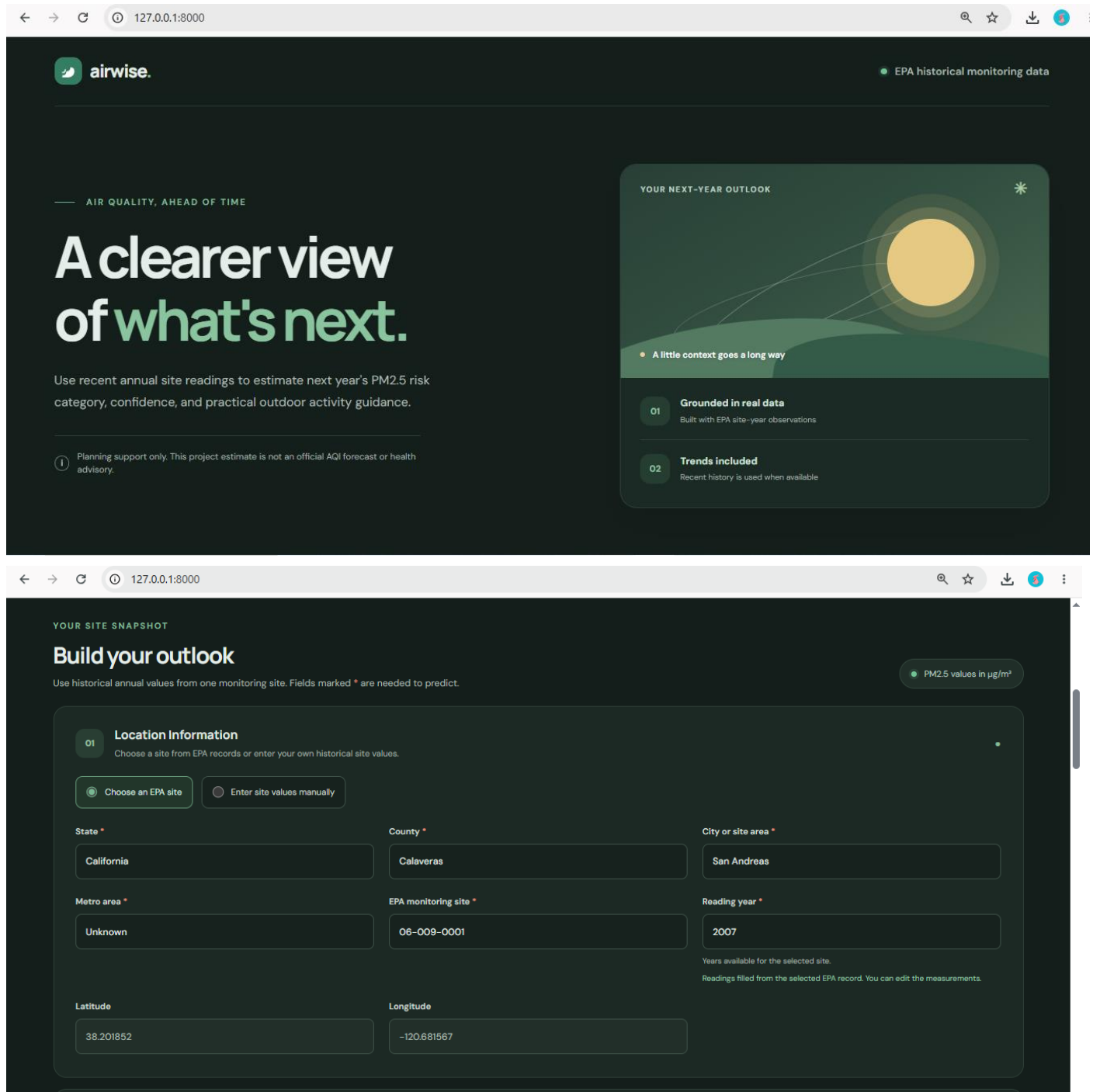


Figure 9. Frontend Home / Site Selection Interface

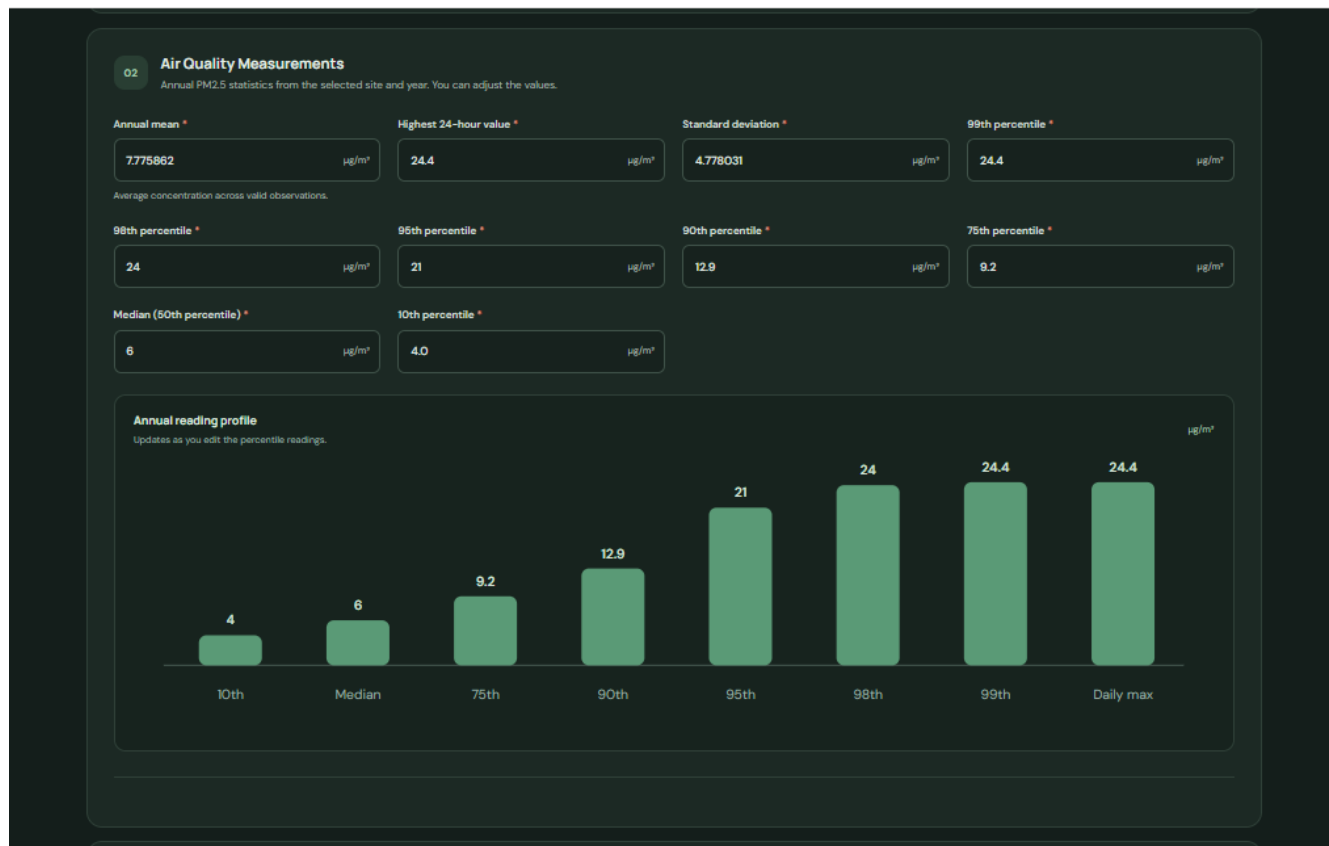


Figure 10. Manual Input / Air-Quality Measurements

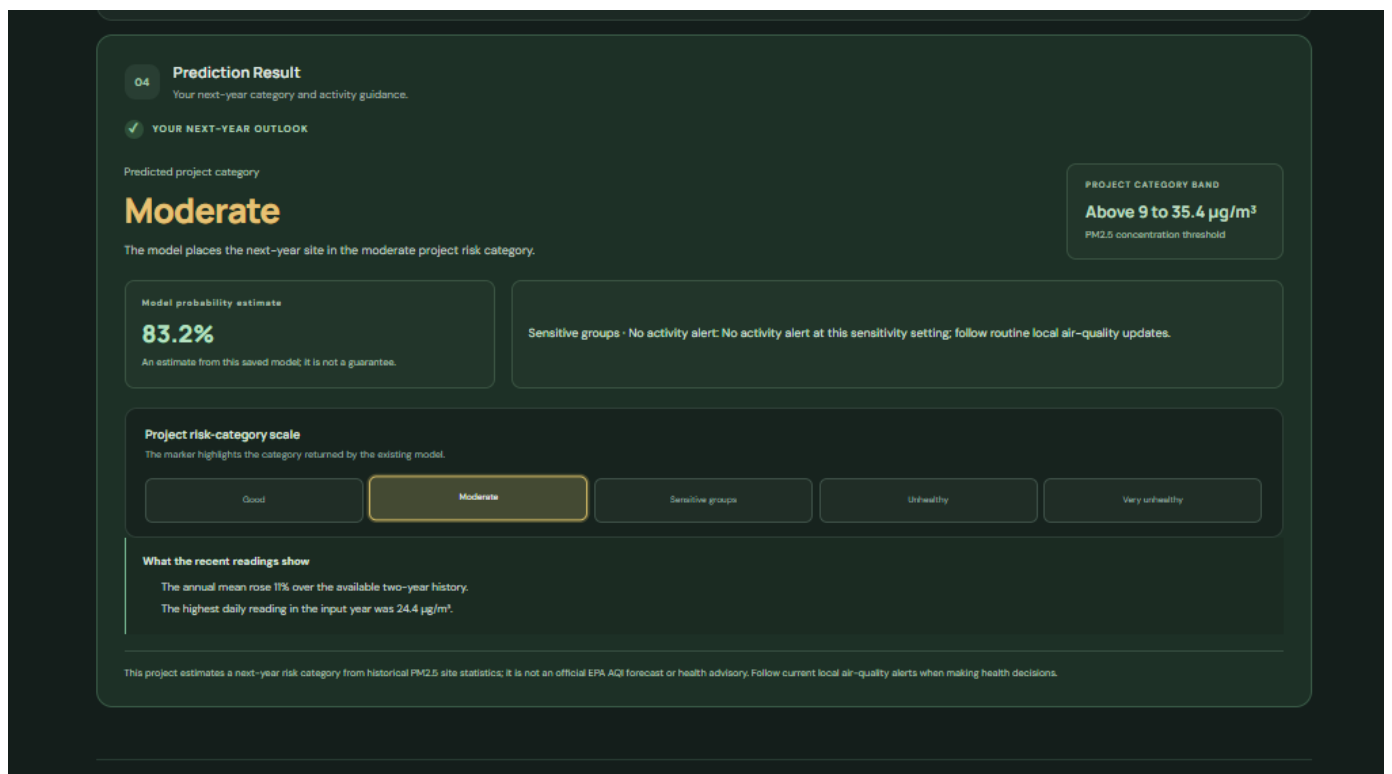


Figure 11. Prediction Result and User Guidance

03 Monitoring Information
Review data coverage, add earlier readings, and choose an alert sensitivity.

Valid observations * days Data coverage * % Valid days in the year * days

Add recent history
Optional — Improves the context when you have it. [Add details](#) ✕

Enter the annual PM2.5 mean from this same site in either or both earlier calendar years. Leave unknown years blank; the model will mark that history as unavailable.

Previous calendar year $\mu\text{g}/\text{m}^3$ Two years before $\mu\text{g}/\text{m}^3$

Alert setting

This changes the activity alert, not the model's predicted category.

Regional overview Predict up to 200 monitoring sites from a CSV ✕

Monitoring site data (CSV)
 No file chosen Regional alert setting

Each row should contain one site's annual statistics. Invalid rows will be listed separately.

[Download CSV template](#) [Predict all sites](#) ↗

🔍 Your entries are used for this prediction only. [Get my outlook](#) ↗

Figure 12. Batch CSV Prediction Interface / Result

Together, these screenshots provide visual evidence of the frontend workflow and the end-to-end integration between user input, backend prediction, and result presentation.

13. System Testing

The repository contains automated tests for request validation, model loading, feature-schema consistency, prediction behavior, batch row limits, CSV parsing, sensitivity guidance, and HTTP endpoints. In the final report verification run, 37 tests passed.

Test Area	Evidence
Input validation	Rejects invalid numeric ranges, missing/invalid fields and malformed CSV headers.
Model integration	Checks saved-model loading and feature names against the fitted schema.
Prediction API	Confirms endpoint returns a model result for valid input.
Batch prediction	Checks per-row errors and maximum row limit.
Sensitivity logic	Confirms alert thresholds change guidance rather than predicted class.
Final verification	37 automated tests passed.

A compatibility warning was observed when loading the submitted model in a scikit-learn 1.8.0 environment because the model had been serialized under scikit-learn 1.9.1. The tests still passed, but deployment should pin the training/runtime library versions to eliminate model-persistence compatibility risk.

14. Limitations and Future Improvements

- The dataset is historical and does not represent regulatory or pollution pattern changes after the source period.
- Severe class imbalance remains the main modelling limitation. The final test set contains only four Very Unhealthy cases and 29 Good cases.
- Temporal distribution shift is substantial: Moderate represents about 51.92% of training records but about 82.45% of test records.
- High accuracy is therefore not sufficient evidence of balanced performance; minority-class recall remains low.
- The engineered categories are project indicators and must not be presented as an official EPA AQI forecast.
- The repository does not preserve the exact original BigQuery extraction query/project/table information, so the extraction step is not fully reproducible from the repository alone.
- Model serialization should use pinned scikit-learn versions or a controlled environment.
- Future work should add newer EPA data, investigate cost-sensitive/ordinal learning, temporal cross-validation, probability calibration, class-aware resampling where defensible, and monitoring for distribution drift.
- Additional interpretability methods could provide clearer per-prediction explanations, especially for public-health stakeholders.

15. Individual and Group Contributions

The repository documents responsibilities by project stage. Where a repository document does not explicitly map a named student to a numbered member, this report avoids inventing an attribution. The team should confirm the final member-to-student mapping before submission if the lecturer requires named individual contributions.

Student ID	Student Name	Individual Contribution
IT23709966	Edirisingha E A P K	Responsible for data loading, data validation, and data cleaning during the data preparation stage. In the modelling stage, developed and evaluated the Random Forest and Gradient Boosting models and performed hyperparameter tuning for both models.
IT23714434	Rajasekara L.G.P.B	Responsible for target engineering and feature engineering, including the preparation of the engineered PM2.5 risk target and predictive features. In the modelling stage, developed and evaluated the LightGBM model and performed its hyperparameter tuning.
IT23688490	Fernando A A I	Responsible for data quality analysis, model-ready dataset creation, and Exploratory Data Analysis (EDA). In the modelling stage, developed and evaluated the Multinomial Logistic Regression model and performed hyperparameter tuning. Also carried out the

		final model selection and comparison using the evaluation results from all candidate models.
IT23715974	Marasinghe M.A.P.I	Responsible for developing the data preprocessing pipeline, including the required transformations for numerical and categorical features. In the modelling stage, developed and evaluated the Support Vector Machine (SVM) with RBF kernel and performed hyperparameter tuning for the SVM model.

Group-level integration included reviewing shared preprocessing assumptions, using a common chronological split, comparing models on the same validation criteria, selecting the final SVM using validation Macro-F1, and integrating the selected pipeline into the prediction application.

16. Conclusion

The project successfully implements the complete data mining workflow for next-year PM2.5 risk classification. Starting from a large real-world EPA air-quality dataset, the team produced a leakage-safe model-ready dataset, performed detailed EDA, engineered temporal and distributional features, created a train-only preprocessing pipeline, and compared five classification algorithms.

The tuned RBF SVM was selected because it achieved the strongest validation Macro-F1 (0.4364) under the predefined selection rule. Its final test accuracy was 79.24%, but test Macro-F1 was 0.2945, showing that the rare categories remain difficult under temporal distribution shift. This distinction is important: the system demonstrates a technically complete and functional prediction workflow, while its limitations must be communicated clearly for any real-world interpretation.

The final model is integrated into a user-facing web application with input validation, site-year selection, manual and batch prediction, configurable activity guidance, and automated backend tests. The resulting system meets the academic objective of connecting data understanding, machine learning, optimization, and deployment in one end-to-end solution.

References

- [1] U.S. Environmental Protection Agency (EPA). *Air Quality System (AQS) / AirData annual air-quality summary data*.
- [2] Kaggle. *EPA Air Quality dataset (epa/air-quality)*.
- [3] Scikit-learn documentation: *preprocessing pipelines, classification metrics, Random Forest, Gradient Boosting, Logistic Regression and Support Vector Machines*.
- [4] LightGBM documentation: *gradient boosting framework for tabular machine-learning tasks*.
- [5] IT3051 – *Fundamentals of Data Mining, Mini Project Assignment 2026*.
- [6] DataBytes. *Dataset Proposal: Air Quality Risk Classification Using EPA Historical Monitoring Data, 2026*.

Appendix A – Repository Structure

app/	Frontend and backend prediction application
data/interim/	Intermediate and model-ready datasets
docs/	Methodology/member documentation
models/	Serialized final model pipeline
notebooks/	EDA, preprocessing, baseline, tuning and selection notebooks
reports/	Data-quality, EDA and model-evaluation outputs
src/data/	Loading, validation, cleaning and splitting
src/features/	Target and historical feature engineering
src/preprocessing/	Reusable preprocessing pipeline
tests/	Automated application/model integration tests

Appendix B – Reproduction / Run Instructions

From the project root on Windows:

```
python -m venv venv
```

```
venv\Scripts\activate
```

```
pip install -r requirements.txt
```

```
python app/backend/server.py
```

Then open `http://127.0.0.1:8000` in a browser. The file `models/best_model.joblib` must be present.

For backend tests, run: `python -m pytest tests/test_prediction_app.py`